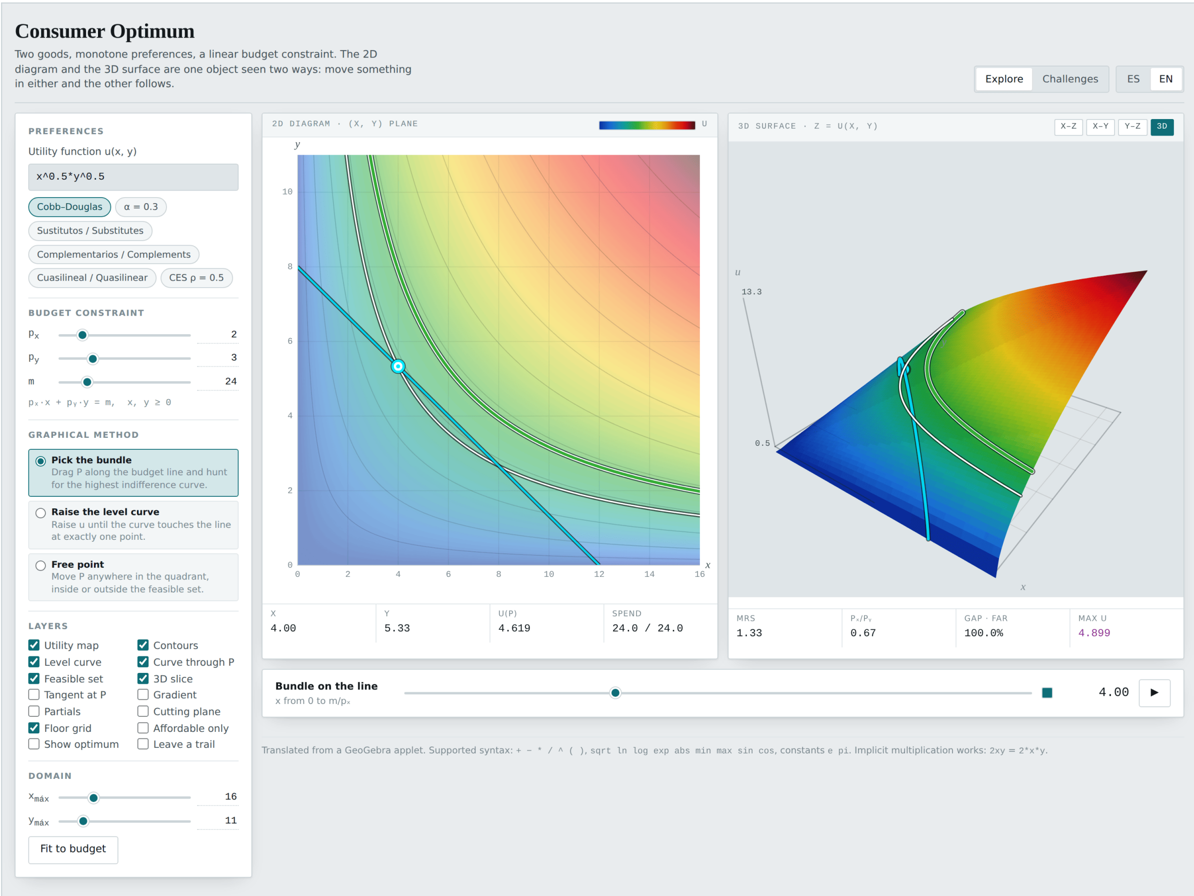
MANUAL

Consumer Optimum

Two goods, monotone preferences and a linear budget constraint. The 2D diagram and the 3D surface are the same thing seen two ways: move something in either and the other follows.



Applet manual · Introductory Microeconomics · Universidad de los Andes
Translated from the GeoGebra applet “Maximización de Utilidad, Curvas de Nivel y. . . ”.
 It opens in a browser; there is nothing to install.

Contents

1 What this is

The consumer’s problem is taught in two dimensions: indifference curves and a budget line. But utility is a function of two variables, and indifference curves are its level curves. The familiar diagram is the view from directly above a surface.

This applet shows both views at once and keeps them coupled. Drag the bundle in the plane and it moves on the surface; drag it on the surface and it moves in the plane. The orthographic X–Z projection turns “find the best affordable bundle” into “find the top of this hill”.

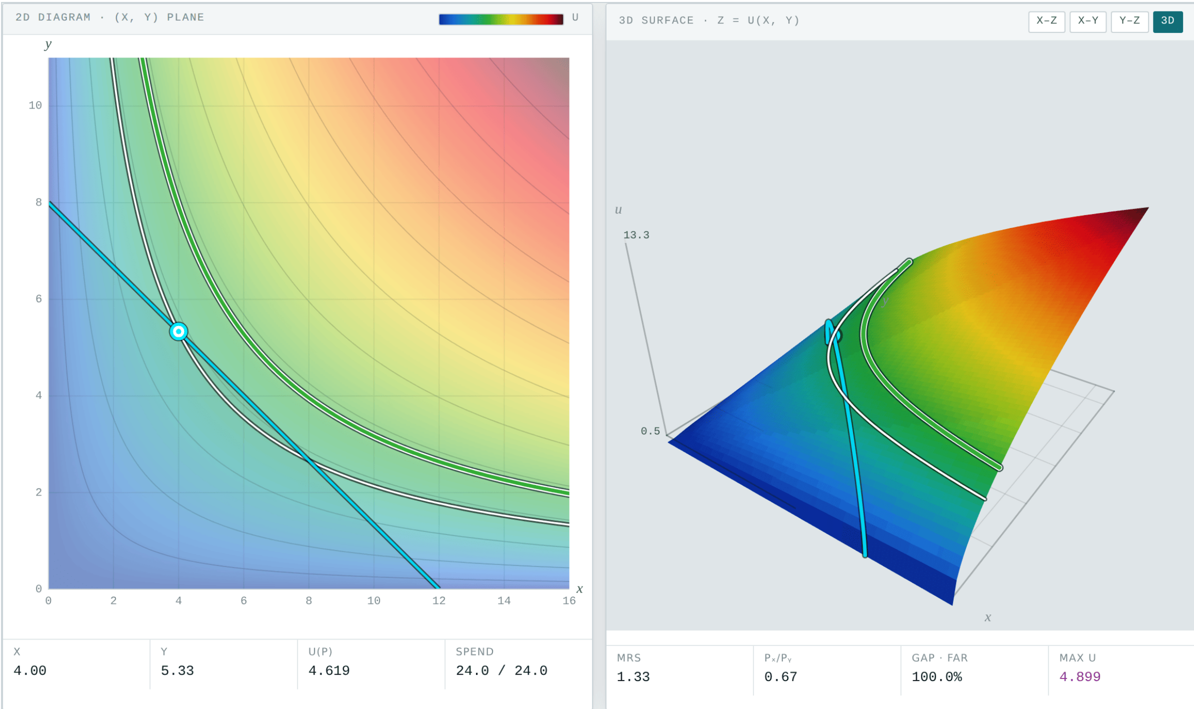


Figure 1. The two panels. Left: the textbook diagram, with the utility map underneath. Right: the same function as a surface $z = u(x, y)$, with the budget line lifted onto it.

The point

The budget curve lifted onto the surface is the object that joins the two views. Its highest point *is* the optimum. Seeing it as a hill with a summit takes the mystery out of the tangency condition.

1.1 Who it is for

The consumer-choice topic, and any course where the level curves of a function of two variables need to stop being an abstract drawing. It also carries a challenge mode with six preference families, scoring and saved progress, which works well as homework.

2 Opening it

There is nothing to install. The whole application is one `index.html` file: it downloads no libraries, calls no server, and stores nothing outside your browser.

2.1 The three ways

Double-click the file. The quickest route. Find `consumer-optimum/index.html` and open it. The browser shows it straight off the disk, with an address beginning `file://`. It works offline from the first moment.

From the web. If someone has published it, the address ends in `/consumer-optimum/`. Once the page has loaded you can disconnect: it needs the network for nothing else.

Served from a folder. For a class, putting the folder on any static server is enough. With Python installed:

```
python3 -m http.server 8000
```

then `http://localhost:8000/consumer-optimum/` in the browser.

> **Note**
>
> Saving the page with `Ctrl` + `S` (or `Cmd` + `S` on a Mac) leaves a copy of your own that still works offline and that can travel on a memory stick or go out by email.

2.2 Language and theme

Two pairs of buttons sit at the top right.

Control	What it does
ES / EN	Switches the whole interface between Spanish and English on the spot. Nothing is recomputed, so you can switch mid-explanation without losing the state you had.
Dark / Light	Switches the theme. Dark is the original design; light exists for projectors and for printing.

Both choices are remembered in the browser, so next time it opens the way you left it.

2.3 Which browser

Any reasonably recent one: Chrome, Edge, Firefox or Safari, on a computer, tablet or phone. On a narrow screen the panels stack one above the other instead of sitting side by side. No account, no permissions, no extensions.

3 The screen

Mode. **Explore** or **Challenges**, top right.

Preferences. The $u(x, y)$ box and six shortcuts.

Budget constraint. p_x, p_y, m.

Graphical method. Which of the three ways of hunting for the optimum is active.

Layers. Fourteen, described below.

Domain. How far the axes run, with a button to fit them to the budget.

Under the two panels sits a slider whose meaning changes with the chosen method, and at the foot of each panel a row of readouts.

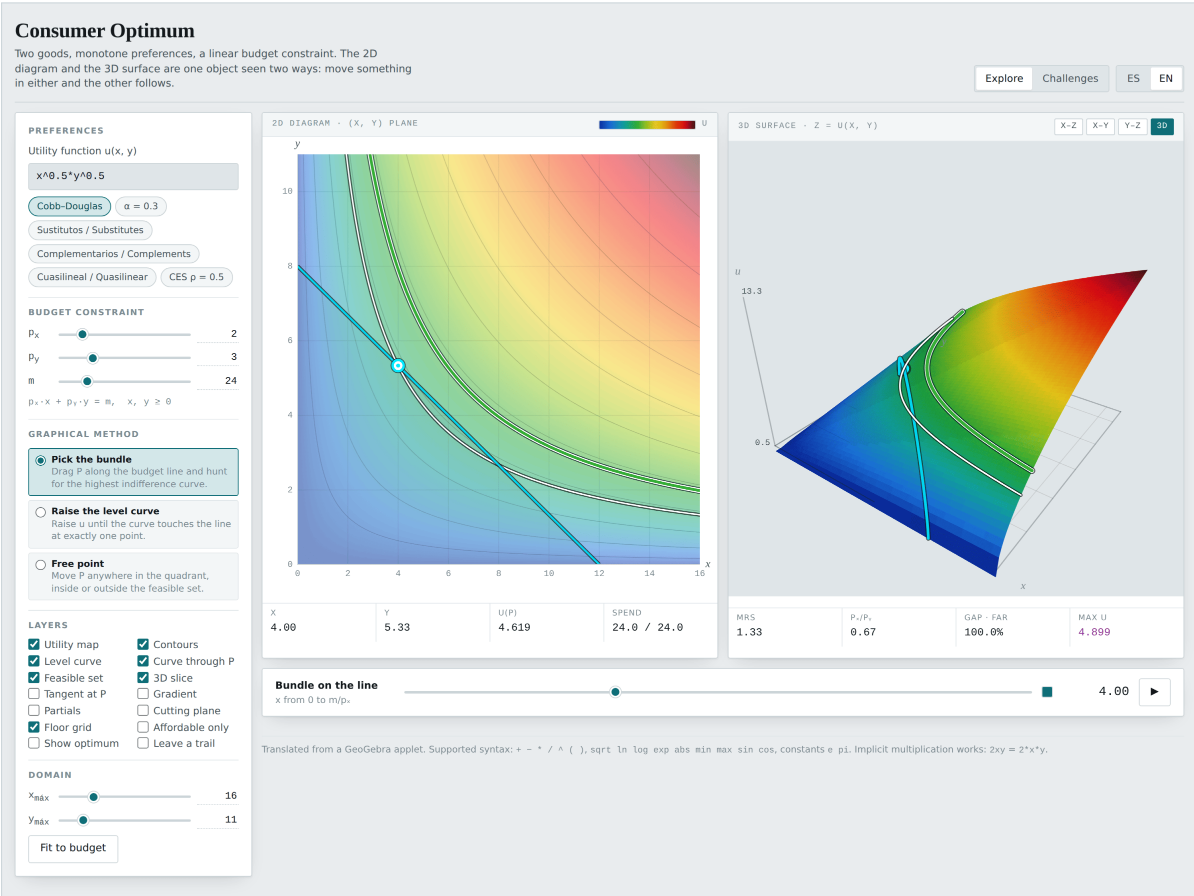


Figure 2. The whole screen.

3.1 The readouts

Readout	What it is
x, y	The coordinates of the bundle P.
$u(P)$	Utility at P.
Spend	What is spent against what is available, $p_x x + p_y y$ against m.
MRS	The marginal rate of substitution at P, u_x/u_y, reported positive.
p_x/p_y	The relative price. At an interior optimum the two agree.
Gap	How far P is from being the optimum, with a label: *tangency*, *near* or *far*.
max u	The utility of the true optimum. The number to reach.

> **Note**
>
> The MRS is reported positive, as u_x/u_y, and compared directly against p_x/p_y. The GeoGebra original displayed $-u_x/u_y$ under a label reading u_x/u_y, which made the two numbers impossible to compare by eye.

4 The three graphical methods

Pick the bundle. P moves along the budget line, and the slider below runs it from $x = 0$ to $x = m/p_x$. The task is to climb while the colour under P keeps getting lighter.

Raise the level curve. P stays put and the slider raises the level u. The task is to find the level at which the curve stops cutting the line twice and stops cutting it at all: there it grazes at a single point.

Free point. P moves anywhere in the quadrant, inside or outside the feasible set. Useful for talking about unaffordable bundles and about ones that leave money unspent.

Both panels are draggable under all three methods. In 3D the pointer is ray-marched into the height field, so the bundle lands where the cursor visually touches the terrain.

5 The camera

Four buttons above the 3D panel. Three are true **orthographic** projections — no perspective divide — so parallel edges stay parallel and equal world steps are equal screen steps.

Button	Projects onto, and which axes end up on screen
X–Z	The plane $y = 0$: x right, u up.
X–Y	The plane $z = 0$: x right, y up.
Y–Z	The plane $x = 0$: y right, u up.
3D	None: free perspective, with orbit and zoom.

X–Z is the one to reach for: "find the best affordable bundle" becomes "find the top of this hill". **X–Y** reproduces the 2D diagram exactly, which makes the correspondence between the two panels concrete. In both side views an indifference curve, being a level set, projects to a horizontal line.

Dragging or arrow-keying away from a canonical view releases the button but keeps the projection type. Dragging the bundle works in every view: the app uses only the coordinates the active projection can express, so a horizontal drag in **Y–Z** moves along y rather than along the collapsed

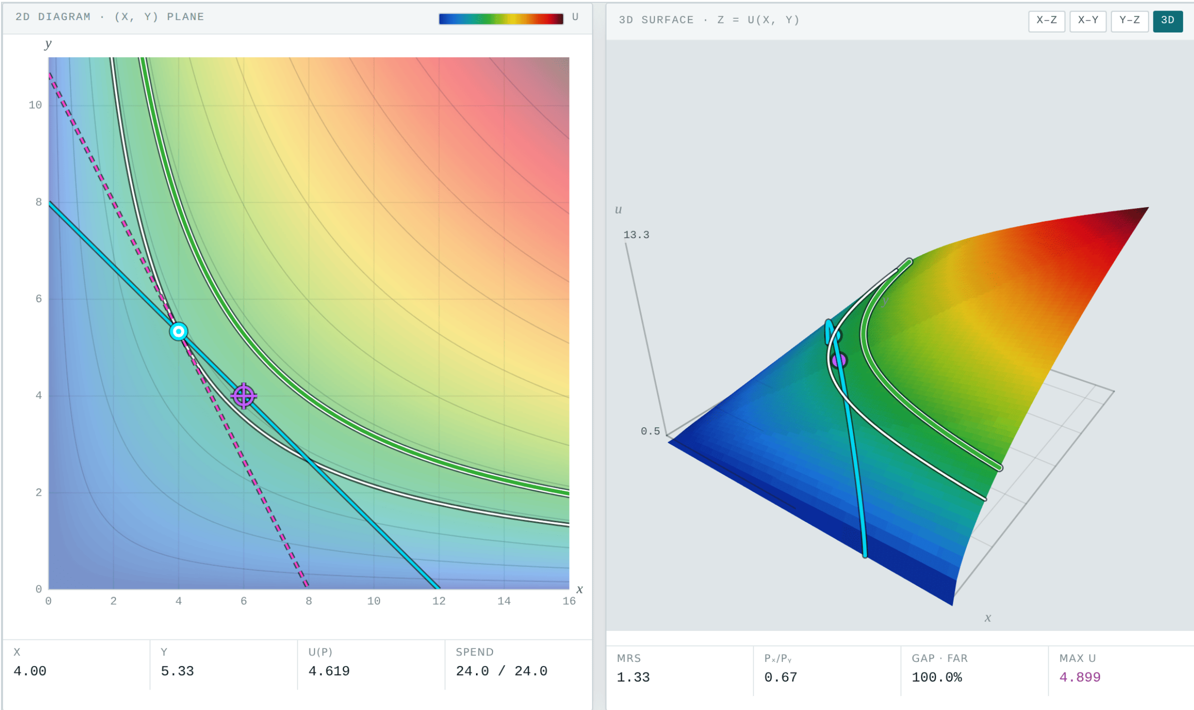


Figure 3. With the **Show optimum** layer and the tangent switched on.

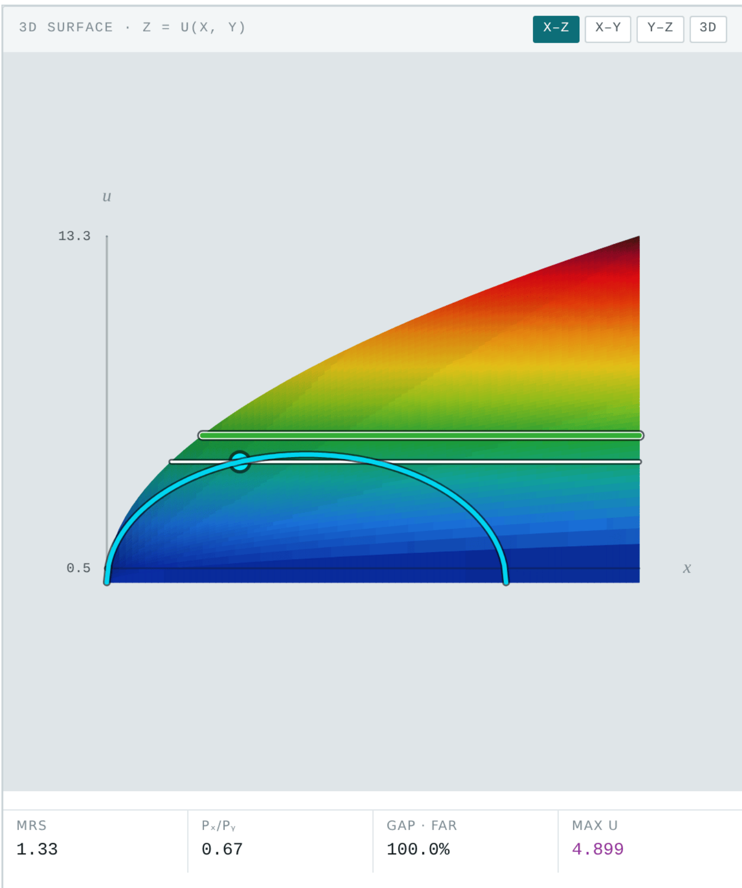


Figure 4. The X–Z projection. The lifted budget curve projects to a single arch whose peak is the optimum.

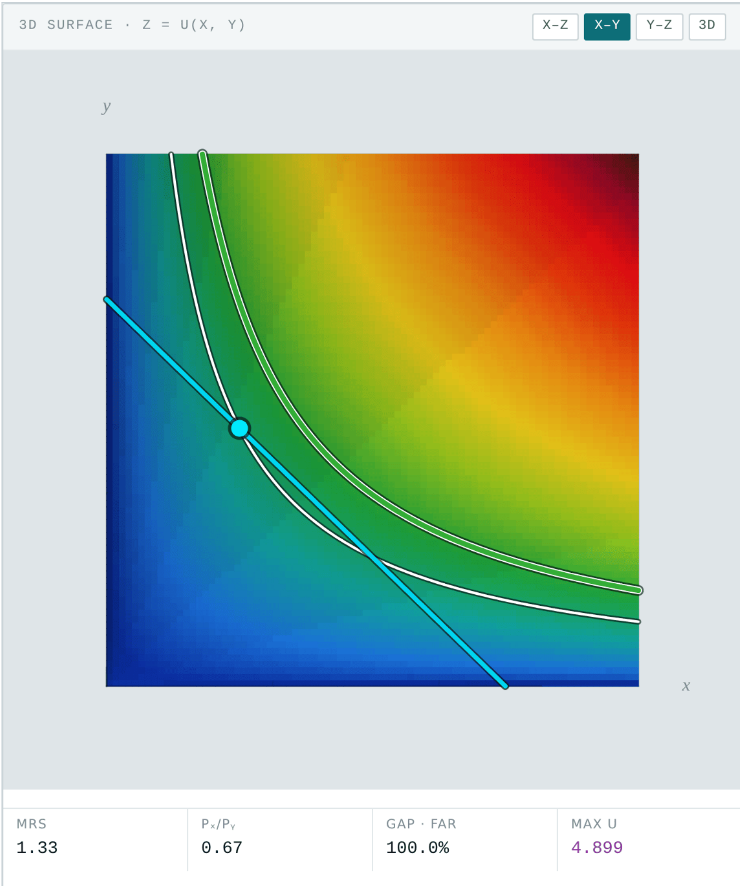


Figure 5. The X–Y projection. It is the 2D diagram, drawn by the 3D engine.

x axis.

6 The layers

Layer	What it draws
Utility map	The field u under a colour ramp.
Background contours	Level curves at regular intervals.
Level curve	The curve at the level the slider is set to.
Curve through P	The indifference curve through the bundle.
Feasible set	Veils what cannot be afforded.
3D slice	The budget line lifted onto the surface.
Tangent at P	The tangent to the indifference curve at P.
Gradient	∇u at P.
Partials	u_x and u_y separately.
Cutting plane	The vertical plane over the budget line.
Floor grid	The ground grid in 3D.
Affordable only	Clips the surface to the feasible set.
Show optimum	Reveals the true optimum.
Leave a trail	Keeps the level curves swept so far.

7 Special cases

The six preference shortcuts cover the cases a course needs. Two are worth looking at slowly, because they break the tangency condition.

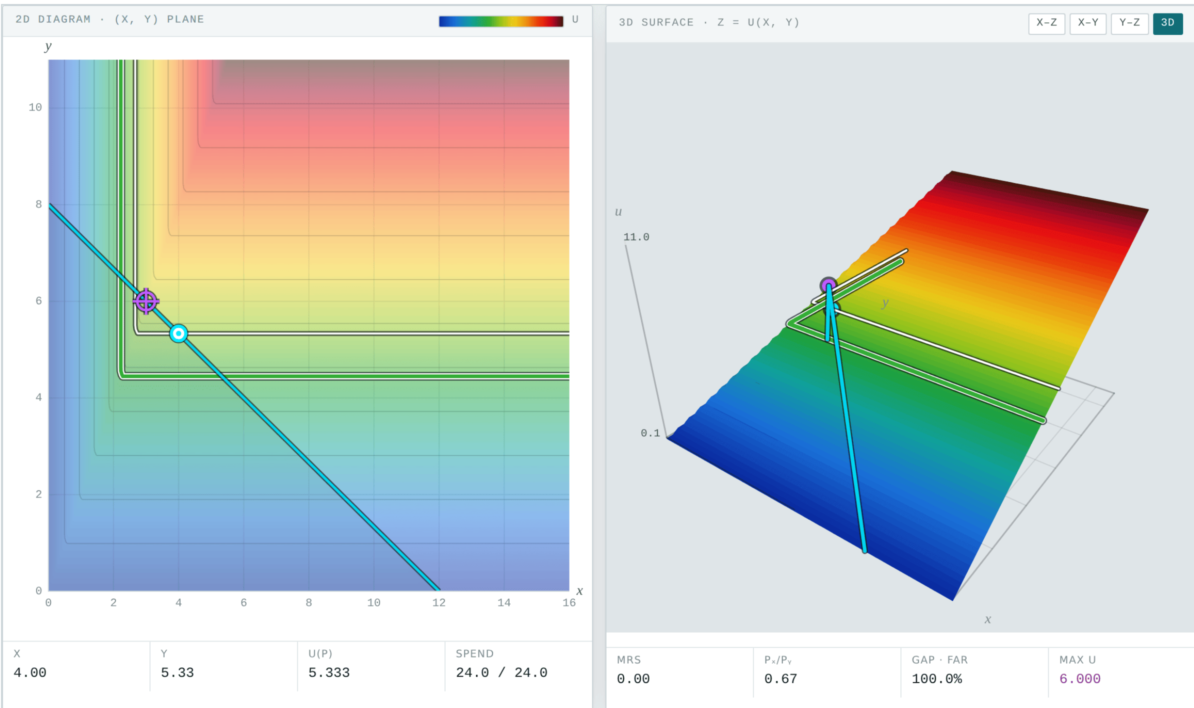


Figure 6. Perfect complements, `min(2x,y)`. The indifference curve has an elbow and there is no tangent to match.

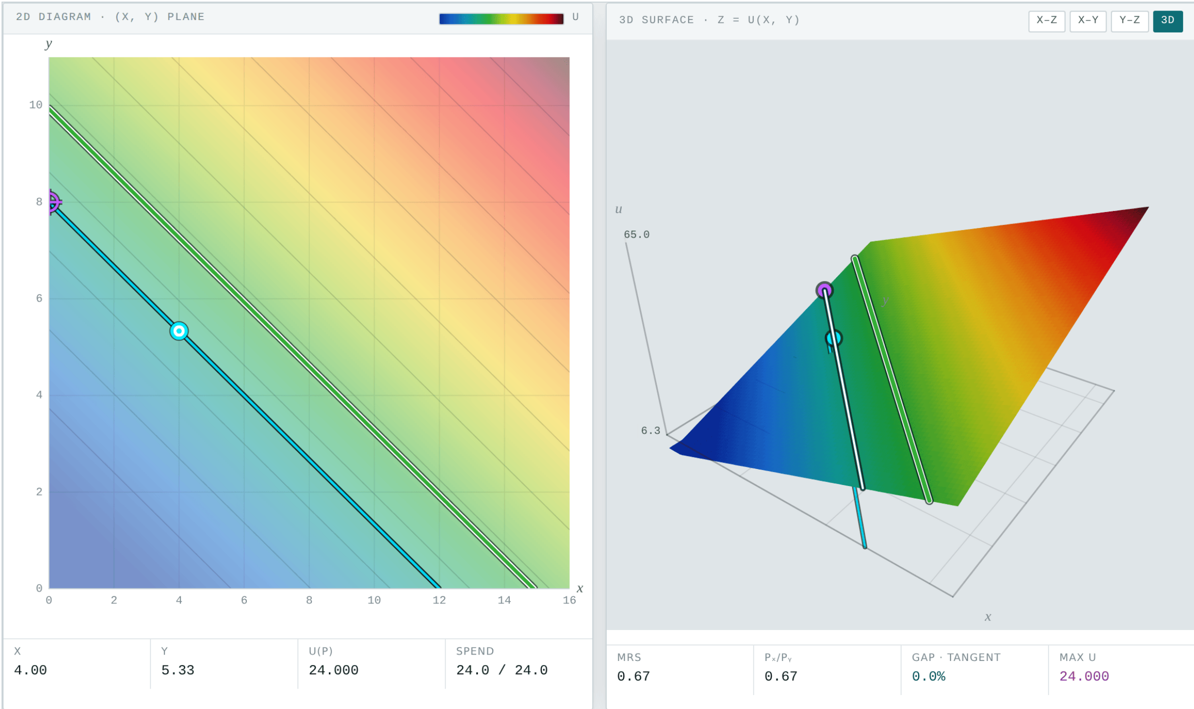


Figure 7. Perfect substitutes, `2x+3y`. With $p_x/p_y = 2/3$ the budget line and the indifference curves are parallel: the whole line is optimal.

> **Note**
>
> At the opening prices ($p_x = 2$, $p_y = 3$) the `2x+3y` shortcut gives exactly MRS $= p_x/p_y$. It is the degenerate case where the consumer is indifferent between every affordable bundle, and it is worth showing before it turns up unannounced in an exam.

8 Challenge mode

Six levels, each with its own preference family and with prices and income generated afresh on every attempt:

Level	Family and task
1	Cobb–Douglas. Place *P* on the optimal bundle.
2	Asymmetric Cobb–Douglas. Raise the level curve until it grazes.
3	Perfect substitutes. Place *P*.
4	Perfect complements. Place *P*.
5	Quasilinear. Raise the level curve.
6	CES. Place *P*.

While a challenge is live, prices, income and the utility function are locked. Checking an answer reveals the optimum, scores the attempt, and explains *which* optimality condition applied: tangency, a kink, or a corner. Progress is kept in the browser.

> **Note**
>
> Alternating the task between “place the bundle” and “raise the curve” is deliberate: they are the two graphical methods on the syllabus, and a student who only practises one does not recognise the other in an exam.

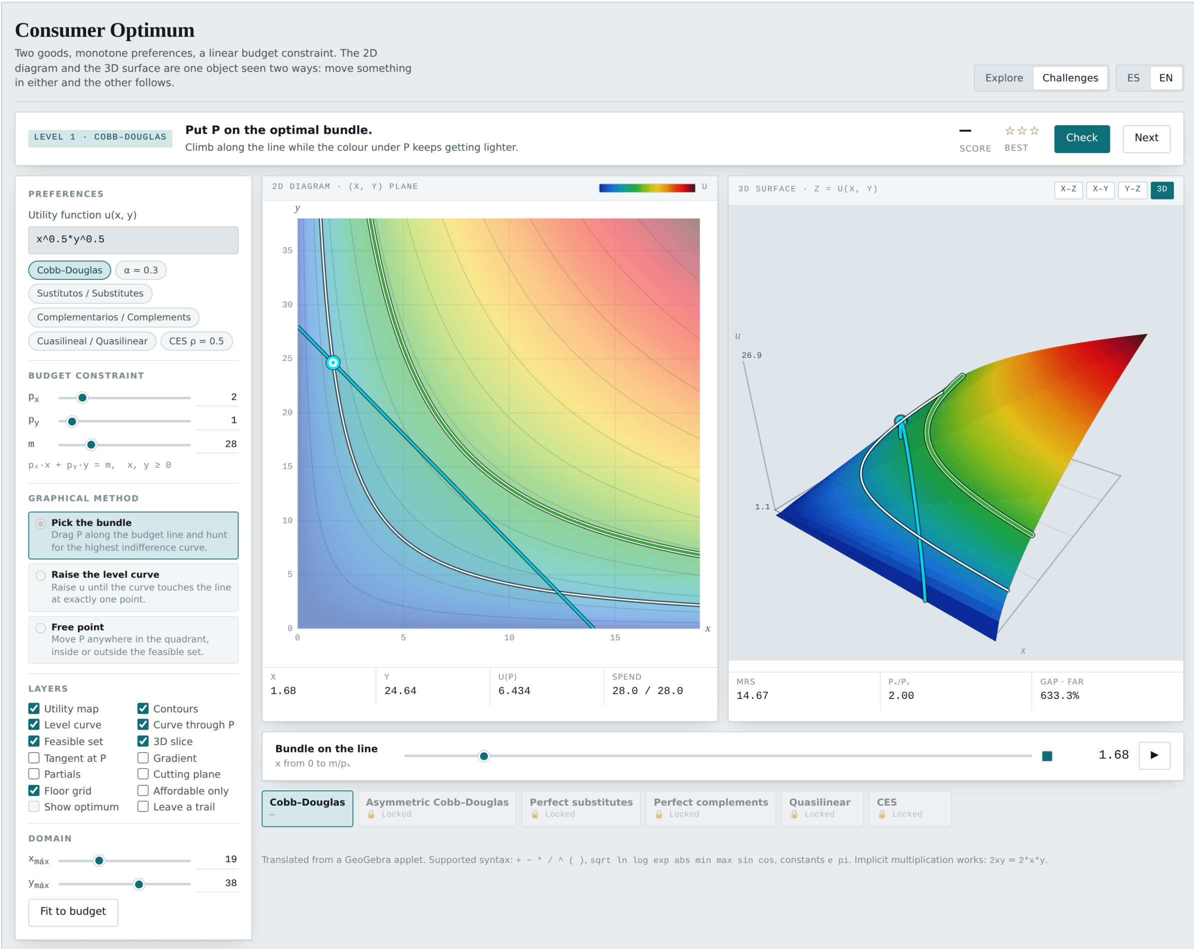


Figure 8. Challenge mode.

9 Classroom activities

Activity 1 · The two views are one

Settings: Opening values ($u = x^{0.5}y^{0.5}$, $p_x = 2$, $p_y = 3$, $m = 24$). Method **Pick the bundle**.

What you should see: Drag P in the 2D panel and watch it move in the 3D one. Then drag it in the 3D panel and watch it move in the 2D one.

Set the camera to **X–Y**: the right panel becomes the left one. It is the shortest proof that an indifference curve is a level curve.

Activity 2 · Finding the summit

Settings: Camera on **X–Z**, method **Pick the bundle**.

What you should see: The lifted budget curve projects to an arch. Move the slider and watch $u(P)$ rise to a maximum and fall again.

The **max u** readout says 4.899, which is $\sqrt{24}$. The optimum is at $x^* = 6$, $y^* = 4$. Check it against the Cobb–Douglas formula $x^* = \alpha m/p_x$ and see that the number matches the top of the arch.

Activity 3 · Tangency, found two ways

Settings: Method **Pick the bundle**; then **Raise the level curve**.

What you should see: With the first method, climb the line while the colour under P keeps lightening: the **Gap** readout falls until it says *tangency*.

With the second, raise u until the curve stops cutting the line twice. The level at which it grazes is the same 4.899. Both syllabus methods give the same number, and the class should see it in the same minute.

Activity 4 · MRS against relative price

Settings: Opening values, **Tangent at P** layer on.

What you should see: With P away from the optimum, **MRS** = 1.33 and $p_x/p_y = 0.67$: the tangent is steeper than the budget line. Move P until the two numbers agree.

There is the first-order condition, read as two numbers chasing each other. Switch on **Gradient** and **Partials** to see where the MRS comes from.

Activity 5 · An optimum with no tangency

Settings: **Complementarios / Complements** shortcut, `min(2x,y)`.

What you should see: The **max u** readout gives 6.000, at $x^* = 3$, $y^* = 6$. But the MRS at the elbow is undefined: the readout gives 0.00 and no tangent matches anything.

Ask how the optimum is found, then. The answer is that you solve $2x = y$ together with the budget line, and the tangency condition plays no part. It is the case that breaks the recipe.

Activity 6 · Every bundle ties

Settings: **Sustitutos / Substitutes** shortcut, `2x+3y`, opening prices.

What you should see: MRS = 2/3 and $p_x/p_y = 2/3$: equal. The budget line and the indifference curves are parallel, so *any* bundle on the line is optimal.

Move p_x slightly either way and the optimum jumps to an axis. That is corner demand, and this is the only setting where you can watch it at the tipping point.

Activity 7 · What cannot be afforded

Settings: Method **Free point**. **Feasible set** layer on.

What you should see: Drag P outside the veil. The **Spend** readout passes 24.0 and the applet does not stop you: the bundle exists, it just cannot be paid for.

Then drag it inwards, leaving money unspent. With monotone preferences that is never optimal, and you can see why: there is always a bundle to the north-east with more colour on it.

Activity 8 · Challenges as homework

Settings: **Challenges** mode.

What you should see: All six levels generate fresh prices and income on every attempt, so the answer cannot be memorised. On checking, the applet explains which condition applied in that case.

It is a one-session assignment that marks itself. Progress is kept in the student's browser, so ask for a screenshot of the score.

10 Expression syntax

Element	What is allowed
Operators	`+ - * / ^ ( )`
Functions	`sqrt ln log log10 exp abs min max sin cos tan`
Constants	`e`, `pi`
Implicit multiplication	`2xy` is `2*x*y`; `x^(1/3)y^(2/3)` parses as expected
Associativity	`^` binds to the right and tighter than unary minus: `-x^2` is $-(x^2)$

11 How it is built

Everything is hand-rolled so the page stays self-contained.

Piece	Approach
Expression language	Tokeniser → Pratt parser → closure tree. No `eval` or `new Function`, so a strict content security policy cannot break it.
Partial derivatives	Symbolic differentiation over the tree, with constant folding. Falls back to central differences when the expression contains `min`, `max` or `abs` — exactly the kink cases.
Level curves	Marching squares over a 141 × 141 sampled field, with saddle disambiguation and NaN-cell skipping.
Optimum	Dense scan then golden-section refinement along the budget line, then classification into interior, kink or corner. Robust where a first-order condition finds nothing.
3D surface	Painter's algorithm on canvas 2D. A height field sorts exactly by centroid depth, so no depth buffer is needed and there is no context-loss failure mode.
3D picking	Ray-march the unprojected pointer ray against the height field, then bisect. Crossings count in both directions, since in a side projection the ray runs level with the terrain and can surface from underneath it.

12 Changes from the GeoGebra original

Colour ramp. The original tinted level curves with three piecewise-linear channel functions (`redfun`, `greenfun`, `bluefun`). The replacement runs blue and green at the bottom, yellow and orange through the middle, then bright red into crimson and deep brown at the top. Because the top end darkens, height is not recoverable from lightness alone up there — the colour bar and the numeric readouts carry that instead, and every curve drawn on the map gets a white casing so it stays legible against the dark browns.

Marks. The bundle, the budget line, the tangent, the gradient, the partials and the revealed optimum are fixed bright hues with a dark casing, none of them reusing a hue the ramp passes through.

Colour range. `uMIN`/`uMAX` were read off two domain corners. For $\ln(x) + y$ that lower corner is $-\infty$ and the ramp collapses. The rewrite takes the 2nd percentile of the sampled field.

Domain. The x_C, y_C sliders reached -10, which puts `sqrt(x)` and `ln(x)` outside their domain. The quadrant is now clamped to $x, y \geq 0$.

Budget parametrisation. x_s ranged over $[x_C, x_G]$ rather than $[0, m/p_x]$, so the "bundle" could sit at negative x with y above the budget line. It is now parametrised by $t \in [0, m/p_x]$.

Prices. p_X and p_Y sliders started at 0, making $y_s = (m - p_X x)/p_Y$ and the domain m/p_X divide by zero. Both are now bounded below by 0.2.

MRS labelling. The original displayed $-f_x/f_y$ under a label reading `RMS := (du/dx)/(du/dy)`. MRS is now reported positive as f_x/f_y and compared directly against p_x/p_y, with the tangency gap as its own readout.

Dead objects. `afin`, `text1` (referring to a_0, a_1, a_2, which do not exist in the file), `BooleFuncion`, `BooleTransparente`, `BooleMostrarVar`, `BooleIntersecciones`, `BooleTangent`, `BooleTanPlane`, `s`, `Impl`, `Boole` and `DX` were leftovers from the applet this was copied from. None were carried over.

13 Accessibility

Both canvases are keyboard-operable: arrow keys move the bundle or the level, and `+` and `-` zoom the 3D view. Light and dark themes are designed separately, and both respond to the system setting. Numbers are tabular. Motion respects `prefers-reduced-motion`.

14 Troubleshooting

Symptom	What is happening
The map is all one colour	The function has a $-\infty$ inside the drawn domain. Raise the domain minimum or change the expression.
I cannot drag in 3D	Check you are not in **Challenges** mode with the controls locked. Otherwise dragging works in all four views.
The surface comes out in pieces	The function is undefined over part of the quadrant. That is correct: there is no ground there.
The camera button switches itself off	You dragged away from the canonical view. The projection type is kept; press it again to recentre.
I lost my challenge score	It is kept in the browser. In a private window, or after clearing site data, it starts from zero.

15 Publishing it for a class

Any static host will do: GitHub Pages, Netlify, a university web directory, even a shared folder served over HTTP. The only thing to upload is the `index.html` file, inside a folder named however you want the address to read.

The repository ships a GitHub Actions workflow (`.github/workflows/pages.yml`) that publishes every applet at once on each push to the default branch. To switch it on, in the repository: *Settings* ⟶ *Pages* ⟶ *Source: GitHub Actions*. That is the one step the workflow cannot do for itself, because creating a Pages site needs administration rights that a `GITHUB_TOKEN` is never granted.

> **Note**
>
> To hand the applet to students without reliable internet, send them the `index.html` file directly. It weighs less than a photograph and opens with a double-click.

16 Licence and provenance

The application code is MIT. Use it, change it and pass it on, in class or out of it, with attribution.

This applet is a standalone rewrite of an original GeoGebra applet. The rewrite is not a literal translation: where the original had a construction error, a division by zero reachable with its own sliders, or a dead object inherited from whatever applet it was copied from, this version fixes it and says so. The section *Changes from the GeoGebra original* lists those changes one by one.

Everything is hand-written and dependency-free: the expression parser, the symbolic differentiation, the level-curve tracing and the drawing. In particular there is no `eval`, so what a student types into a text box cannot become code, and a strict content security policy does not break the page.